Algorithms for 10 Python Case Studies

Comprehensive step-by-step algorithmic documentation for academic reference and project reports.

01 University Scholarship Allocation

1. Start.
2. Input the number of scholarship applicants.
3. For each applicant, input name, aggregate marks, subject-wise marks, attendance percentage, and family income.
4. Check whether aggregate marks are at least 75%, every subject mark is at least 60%, attendance is at least 75%, and family income is at most ₹2,50,000.
5. If all conditions are satisfied, mark the applicant as eligible; otherwise, mark them as ineligible.
6. Sort eligible applicants by higher aggregate marks, higher attendance, and lower family income.
7. Display the ranked eligible applicants and ineligible applicants.
8. Stop.

02 Hospital Billing and Patient Classification

1. Start.
2. Input the patient's name, age, concession-card status, and medical charges.
3. Calculate the total bill by adding all charges.
4. If the patient has a valid concession card, apply a 15% discount.
5. Otherwise, if the patient is a senior citizen aged 60 or above, apply a 10% discount.
6. Otherwise, apply no discount.
7. Calculate the final bill after the discount.
8. Classify the bill as High if it is at least ₹10,000, Medium if it is at least ₹5,000, or Low otherwise.
9. Display the total bill, discount, final bill, and classification.
10. Stop.

03 Multi-Currency Travel Budget Analysis

1. Start.
2. Input the travel budget in Indian rupees.
3. Input travel expenses, their amounts, and their currencies.
4. Convert each expense into Indian rupees using the specified exchange rates.
5. Apply the 2% conversion fee to expenses requiring currency conversion.
6. Calculate the total travel expense.
7. Compare the total expense with the travel budget.
8. Identify the largest expense.
9. Display converted expenses, total cost, budget status, remaining budget or excess amount, and largest expense.
10. Stop.

** Note: The exchange rates used in the program are illustrative.*

04 Campus Event Financial Planning

1. Start.
2. Input the venue, equipment, catering, transport, and other costs.
3. Calculate the total event cost.
4. Input the sponsorship received and subtract it from the total event cost.
5. Calculate the contribution required per participant.
6. Input the target sponsorship and calculate the sponsorship shortfall.
7. Input the registration fee per participant.
8. Use a loop to calculate the number of registrations required to cover the remaining cost.
9. Display the total cost, remaining cost, participant contribution, sponsorship shortfall, and break-even registrations.
10. Stop.

05 Examination Result and Grace-Mark Validation

1. Start.
2. Input the number of subjects, marks obtained, maximum marks, pass mark, and maximum grace marks allowed.
3. Validate the marks for each subject.
4. Check whether the student has passed each subject.
5. If the marks are below the pass mark, determine whether grace marks can bring the student up to the pass mark within the permitted limit.
6. Add the permitted grace marks where applicable.
7. Calculate the total marks and percentage.
8. Determine the final result and classification according to the program's rules.
9. Display subject-wise original marks, grace marks, final marks, total, percentage, and result.
10. Stop.

06 Attendance Recovery Computation

1. Start.
2. Input the number of classes attended and the total classes conducted.
3. Input the required attendance percentage.
4. Calculate current attendance using:

   ```
   Attendance = (Classes Attended / Classes Conducted) × 100
   ```

5. Calculate the additional classes required if the student attends every future class.
6. Calculate the additional classes required if the student attends only 90% of future classes.
7. Display current attendance and both recovery estimates.
8. Stop.

07 ATM Transaction Sequence Simulation

1. Start.
2. Input the opening account balance.
3. Input the number of transactions.
4. For each transaction, input the transaction type and amount.
5. If the transaction is a deposit, check that the amount is positive and add it to the balance.
6. If the transaction is a withdrawal, check that the amount is positive, does not exceed ₹10,000 per

08 Warehouse Inventory Reconciliation

1. Start.
2. Input the item name, opening stock, incoming stock, dispatched stock, recorded closing stock, and minimum stock level.
3. Calculate the expected closing stock:

```
Expected Stock = Opening Stock + Incoming Stock - Dispatched Stock
```

4. Compare expected closing stock with recorded closing stock.
5. Calculate the discrepancy.
6. If both values match, display “Match”; otherwise, display “Mismatch”.
7. Calculate the replenishment quantity required to reach the minimum stock level.
8. Display stock details, discrepancy, stock status, and replenishment quantity.
9. Stop.

09 Smart Parking Fee Management

1. Start.
2. Input the vehicle type, entry time, and exit time.
3. Calculate the parking duration.
4. If the exit time is earlier than the entry time, treat the exit as occurring on the next day.
5. Round the parking duration up to the next whole hour when necessary.
6. Determine the hourly parking rate based on the vehicle type.
7. Calculate the parking fee using the hourly rate and the program's daily-limit rules.
8. Apply the additional penalty if the parking duration exceeds 24 hours.
9. Display the vehicle type, duration, parking fee, penalty, and total fee.
10. Stop.

10 Multi-Day Water Tank Simulation

1. Start.
2. Input the tank capacity, initial water level, target water level, and number of time intervals.
3. For each interval, input the water inflow and outflow.
4. Add the inflow to the current water level.
5. If the available water exceeds the tank capacity, calculate the overflow and limit the water level to the capacity.
6. If the outflow exceeds the available water, calculate the shortage and set the water level to zero.
7. Otherwise, subtract the outflow from the available water.
8. Track the total overflow, total shortage, and first interval when the target water level is reached.
9. Display the final water level and simulation summary.
10. Stop.